Origami 101

University of Toronto, Thursdays 3-5pm, Winter 2025

Abstract. We will study different topics in mathematics via origami. Some of them are related to origami itself as the object of study and others are going to use origami as their inspiration.

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1 Week 3: January 23rd Meeting

1.1 Motivation & Paper Folding

Motivation. *Given a piece of paper & a fold, is the fold flat?*

Suppose I show you an unfolded crease pattern (Figure 1). Are there criterias to decide whether it constructs something or not?

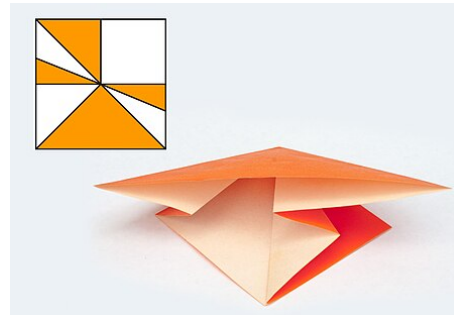

Figure 1: Flat fold and its respective unfolded crease pattern

1.2 Flat-Foldability & Restrictions

Local Restrictions (Necessary): What happens around the interior vertices?

Global Restrictions (Sufficient): Satisfying all local restrictions what remains? i.e. Paper crashing into itself, in Figure 2 having a colour along a column or row of an orthogonal crease pattern is a global restriction

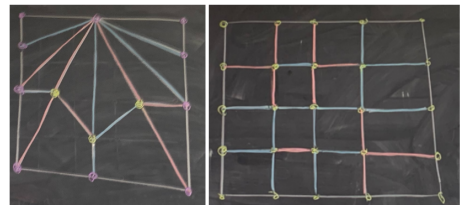

Figure 2: Non-rthogonal & orthogonal crease pattern, blue = mountain crease, red = valley crease

Definition. *If a vertex satisfies all local restrictions it is called locally flat-foldable*

Definition. *If the entire system satisfies all global restrictions it is called globally flat-foldable*

Thus, if a crease pattern is locally & globally flat-foldable the entire system is foldable.

1.3 Vertex Neighborhoods

Each interior vertex has a neighborhood in which it is isolated. See Figure 3.

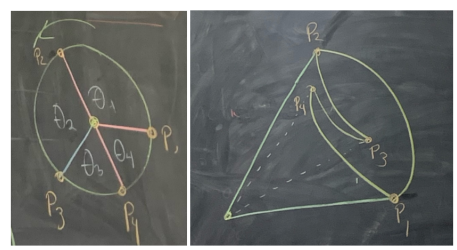

Figure 3: Isolated vertex angles & its flat folded construction from creases $P_1, ..., P_4$

1.4 Local Condition Theorems

Refer to Figure 4 for isolated local vertices.

Theorem 1 (Maekawa). *|Mountain - Valley| = 2*

Theorem 2 (Justin-Kawasaki). *In a flat foldable vertex,* $\theta_1 + \theta_3 + ... = \theta_2 + \theta_4 + ... = 180°$

Also Justin-Kawasaki's theorem also turns out to be general for Riemannian manifolds, see Figure 4

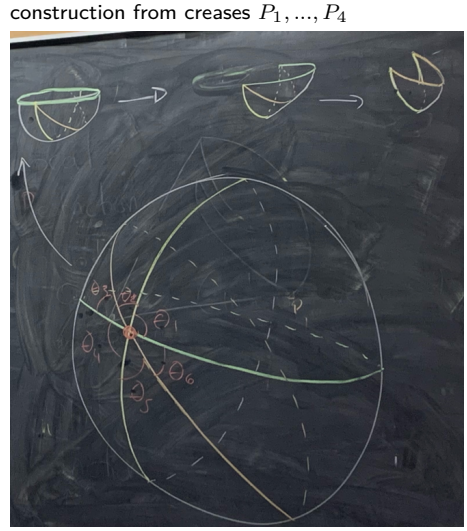

Figure 4: Folding a sphere

2 Week 2: January 16th Meeting

This week was on geometries to motivate an application to origami criterions coming in the following weeks.

2.1 Geometries

Theorem 3. *Gauss's Theorema Egregium "Remarkable Theorem" concerns the curvature of surfaces (i.e. Gaussian curvature is a local isometric invariant)*

In classical geometry we have the following Euclid's axioms,

1. To draw a straight line from any point to any point.
2. Through any point & with any point as radius there is a circle.
3. All right angles are equal to each other.
4. You can extend a finite straight line continuously in a straight line indefinitely.

5. Through any point not in a lines there is a unique parallel to the line through the point.

Figure 5 show a Gauss map N and its derivative.

Theorem 4 (Gauss-Bonnet)**.**

$$\int_S K(t)dt = 2\pi X(S)$$

Where $K(t)$ is the Gaussian curvature and $X(S)$ is the Euler characteristic $v - \epsilon + f$ (vertices-edges+faces) which is equal to 2 for a sphere and 0 for a torus

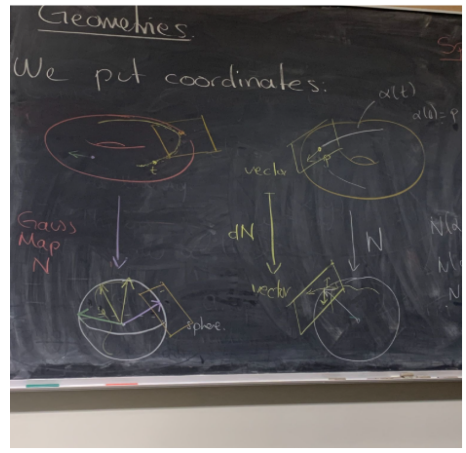

Figure 5: Gauss map N and its derivative dN

References

[1] Robert J. Lang. *Twists, Tilings and Tessellations. Mathematical Methods for Geometric Origami*. CRC Press, 2017.

[2] Erik D. Demaine and Joseph O'Rourke. *Geometric Folding Algorithms. Linkages, Origami, Polyhedra*. Cambridge University Press, 2007.

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[4] Joseph O'Rourke. *How to fold it. The Mathematics of Linkages, Origami and Polyhedra*. Cambridge University Press, 2011.

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[6] Robert J. Lang. *Origami 6 Sixth International Meeting of Origami Science, Mathematics and Education*. CRC Press, 2015.